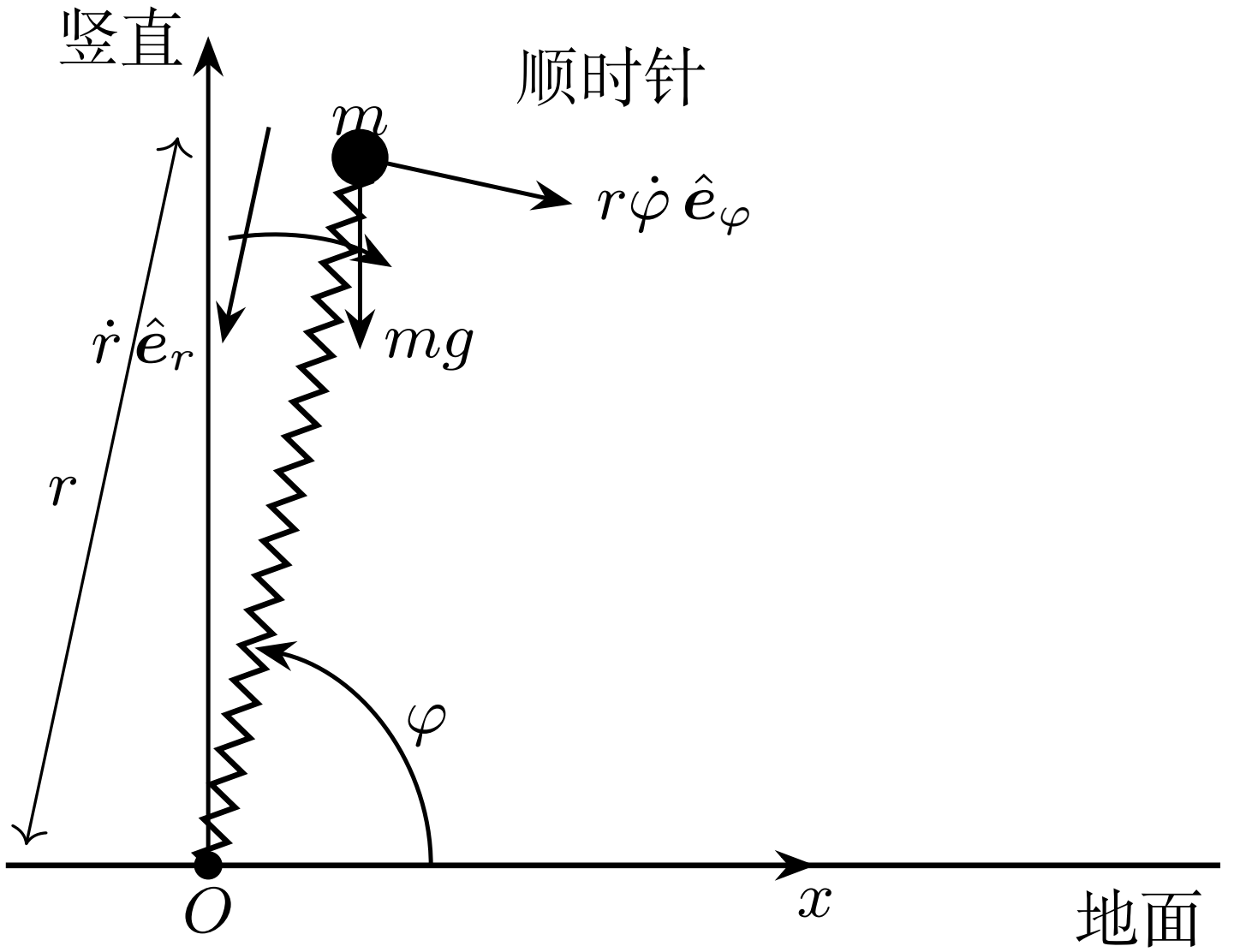
竖直
顺时针
m
$r\dot{\varphi}\,\hat{\boldsymbol{e}}_\varphi$
$\dot{r}\,\hat{\boldsymbol{e}}_r$
mg
r
φ
O
x
地面